# High-symmetry points and paths in the Brillouin zone

and what they dictate for the LiF polaron dispersion

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**Abstract**

The polaron-dispersion campaign measures $E_0(\mathbf{p})$ at nonzero external momentum and extracts the effective mass $m^*$ from its curvature at $\Gamma$. Choosing *which* momenta to run means choosing directions in the Brillouin zone, and the conventional choices — "$\Gamma \to X$", "$\Gamma \to L$" — come from crystal symmetry. This note defines high-symmetry points and paths from scratch for the fcc lattice of LiF, maps them onto the fractional-coordinate $n^3$ grid the sampler uses, derives what cubic symmetry forces on the small-$\mathbf{p}$ expansion of $E_0(\mathbf{p})$ (an isotropic scalar $m^*$ for the non-degenerate conduction band, direction dependence only at quartic order), and closes with an annotated reading list. The methodology note (`latex/ab-initio-diagmc`) remains the authoritative reference for the sampler and estimators; this note only covers the symmetry background.

## 1 Reciprocal lattice, Brillouin zone, and fractional coordinates

LiF crystallizes in the rock-salt structure: a face-centered-cubic (fcc) Bravais lattice with a two-atom basis. With $a$ the conventional cubic lattice constant, the primitive real-space vectors are

$$\mathbf{a}_1 = \tfrac{a}{2}(0,1,1), \qquad \mathbf{a}_2 = \tfrac{a}{2}(1,0,1), \qquad \mathbf{a}_3 = \tfrac{a}{2}(1,1,0), \tag{1}$$

and the reciprocal primitive vectors, fixed by $\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$, are

$$\mathbf{b}_1 = \tfrac{2\pi}{a}(-1,1,1), \qquad \mathbf{b}_2 = \tfrac{2\pi}{a}(1,-1,1), \qquad \mathbf{b}_3 = \tfrac{2\pi}{a}(1,1,-1). \tag{2}$$

The first Brillouin zone (BZ) — the Wigner–Seitz cell of the reciprocal lattice — is a truncated octahedron: eight regular hexagonal faces (perpendicular to the cube diagonals $\langle 111 \rangle$) and six square faces (perpendicular to the cube axes $\langle 100 \rangle$).

Throughout the codebase momenta are stored in *fractional (crystal) coordinates* $\mathbf{f} = (f_1, f_2, f_3)$, meaning

$$\mathbf{k} \;=\; f_1\mathbf{b}_1 + f_2\mathbf{b}_2 + f_3\mathbf{b}_3, \tag{3}$$

on a $\Gamma$-centered $n^3$ grid $f_i = j_i/n$, $j_i \in \{0, \ldots, n-1\}$. Two consequences matter for everything below. First, the grid is closed under addition modulo reciprocal-lattice vectors, which is what keeps every diagram's momentum bookkeeping on-grid. Second, the map (3) is *not* an isometry — the $\mathbf{b}_i$ are neither orthogonal nor unit-length — so equal steps in fractional coordinates are not equal steps in physical momentum, and any fit in $|\mathbf{p}|$ must first convert to Cartesian via (3).

## 2 High-symmetry points

The point group of the crystal (for rock-salt: the full cubic group $O_h$, 48 operations — rotations, mirrors, inversion) acts on reciprocal space and maps the BZ onto itself. For a given $\mathbf{k}$, the subgroup of operations that leave $\mathbf{k}$ invariant *modulo a reciprocal-lattice vector* is called the *little group* (or stabilizer) of $\mathbf{k}$. A generic $\mathbf{k}$ has a trivial little group; points whose little group is larger are called *high-symmetry points* and carry conventional letters. For the fcc BZ:

| Point | Fractional $(f_1, f_2, f_3)$ | Cartesian $[\frac{2\pi}{a}]$ | Little group | Location |
|---|---|---|---|---|
| $\Gamma$ | $(0,0,0)$ | $(0,0,0)$ | $O_h$ (48) | zone center |
| $X$ | $(\frac{1}{2},0,\frac{1}{2})$ | $(0,1,0)$ | $D_{4h}$ (16) | square-face center, $\langle 100\rangle$ |
| $L$ | $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$ | $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$ | $D_{3d}$ (12) | hexagon center, $\langle 111\rangle$ |
| $W$ | $(\frac{1}{2},\frac{1}{4},\frac{3}{4})$ | $(\frac{1}{2},1,0)$ | $D_{2d}$ (8) | square–hexagon–hexagon corner |
| $K$ | $(\frac{3}{8},\frac{3}{8},\frac{3}{4})$ | $(\frac{3}{4},\frac{3}{4},0)$ | $C_{2v}$ (4) | hexagon–hexagon edge midpoint |
| $U$ | $(\frac{5}{8},\frac{1}{4},\frac{5}{8})$ | $(\frac{1}{4},1,\frac{1}{4})$ | $C_{2v}$ (4) | square–hexagon edge midpoint |

(Orders of the little groups in parentheses. The Cartesian columns follow from (3) with (2); e.g. for $X$: $\frac{1}{2}\mathbf{b}_1 + \frac{1}{2}\mathbf{b}_3 = \frac{2\pi}{a}(0,1,0)$.) The set of all images of a point under the point group is its *star*; e.g. the $X$ star has the 6 members $\frac{2\pi}{a}(\pm1,0,0), (0,\pm1,0), (0,0,\pm1)$, and the $W$ star (24 members) already appears in the glossary because bands 6/7 are degenerate on it.

Why these points matter physically: any function with the lattice symmetry — $\varepsilon(\mathbf{k})$, $\omega_\nu(\mathbf{q})$, and the polaron dispersion $E_0(\mathbf{p})$ alike — satisfies $E(g\mathbf{k}) = E(\mathbf{k})$ for every point-group operation $g$. At a high-symmetry point the gradient components along symmetry-broken directions must vanish, so band extrema, degeneracies, and critical points concentrate there. The conduction-band minimum of LiF sits at $\Gamma$; the polaron ground state inherits that.

## 3 Symmetry paths

A *symmetry path* is a piecewise-straight tour through high-symmetry points, along edges and face normals of the irreducible BZ wedge. Band structures are functions on a three-dimensional domain; the community convention is to plot them along a standard 1D path so that any two calculations are visually comparable. For fcc the standard tour (Setyawan–Curtarolo convention) is

$$\Gamma \to X \to W \to K \to \Gamma \to L \to U \to W \to L \to K \,\big|\, U \to X,$$

where "|" marks a discontinuous jump. Nothing is physically special about the path as a whole — it is a plotting convention — but the individual *legs* from $\Gamma$ are special: they are the extremal directions along which effective-mass expansions are conventionally quoted, and symmetry constrains $E(\mathbf{k})$ most strongly there. For the dispersion campaign only the first legs out of $\Gamma$ matter: $\Gamma \to X$ (a $\langle 100\rangle$ cube axis) and $\Gamma \to L$ (a $\langle 111\rangle$ cube diagonal).

## 4 The grid rays the sampler can run

The external momentum `p_ext` must be a point of the $n^3$ grid, so a "path" for us is a ray of grid points. The two legs from $\Gamma$:

$$\Gamma \to X: \quad \mathbf{f}_j = \tfrac{j}{n}(1,0,1) \;\Rightarrow\; \mathbf{p}_j = \tfrac{2\pi}{a}\,\tfrac{2j}{n}\,(0,1,0), \qquad |\mathbf{p}_j| = \tfrac{4\pi}{a}\,\tfrac{j}{n}, \qquad j = 0,\ldots,\tfrac{n}{2}, \tag{4}$$

$$\Gamma \to L: \quad \mathbf{f}_j = \tfrac{j}{n}(1,1,1) \;\Rightarrow\; \mathbf{p}_j = \tfrac{2\pi}{a}\,\tfrac{j}{n}\,(1,1,1), \qquad |\mathbf{p}_j| = \tfrac{2\pi\sqrt{3}}{a}\,\tfrac{j}{n}, \qquad j = 0,\ldots,\tfrac{n}{2}. \tag{5}$$

Both reach their endpoint exactly at $j = n/2$ (all ladder grids are even). Note the change of appearance under the fractional→Cartesian map: the "diagonal-looking" fractional ray $(j/n, 0, j/n)$ is a Cartesian *cube axis*, and the fractional body diagonal is the Cartesian cube diagonal. The Cartesian step sizes differ between the legs ($\frac{4\pi}{a}\frac{1}{n}$ vs $\frac{2\pi\sqrt{3}}{a}\frac{1}{n}$), which the fit runbook must account for when comparing the two directions at matched $|\mathbf{p}|$.

## 5 What cubic symmetry dictates for $E_0(\mathbf{p})$ and $m^*$

**Assumptions, stated explicitly.** (i) The state being followed is *non-degenerate* at $\Gamma$ (true for the LiF conduction-band minimum, a single Li-$s$-like band, and hence for the electron polaron built on it; the coupling $g(\mathbf{k}, \mathbf{q}, \nu)$ respects the crystal symmetry, so the interacting dispersion inherits $O_h$). (ii) $\mathbf{p}$ stays below the phonon-emission threshold (below), so $E_0(\mathbf{p})$ is a sharp quasiparticle branch. (iii) Inversion symmetry (contained in $O_h$) plus time reversal give $E_0(\mathbf{p}) = E_0(-\mathbf{p})$, so the expansion is even.

Under these assumptions the most general small-$\mathbf{p}$ expansion invariant under the cubic group is

$$E_0(\mathbf{p}) = E_0 + \frac{p_x^2 + p_y^2 + p_z^2}{2m^*} + A\left(p_x^4 + p_y^4 + p_z^4\right) + B\left(p_x^2 + p_y^2 + p_z^2\right)^2 + \mathcal{O}(p^6). \tag{6}$$

The structure is forced by invariant theory: the only symmetric rank-2 tensor invariant under $O_h$ is $\propto \delta_{ij}$ — so the quadratic term is *isotropic*, a single scalar $m^*$ with no direction dependence — while at quartic order there are exactly two independent invariants, $\sum_i p_i^4$ and $|\mathbf{p}|^4$. Restricted to the two legs:

$$E_0(\mathbf{p}) - E_0 = \frac{|\mathbf{p}|^2}{2m^*} + \begin{cases} \left(A + B\right)|\mathbf{p}|^4 + \ldots & \text{along } \Gamma \to X \ (\langle 100\rangle), \\ \left(\frac{A}{3} + B\right)|\mathbf{p}|^4 + \ldots & \text{along } \Gamma \to L \ (\langle 111\rangle), \end{cases} \tag{7}$$

using $\sum_i p_i^4 = |\mathbf{p}|^4$ on $\langle 100\rangle$ and $= |\mathbf{p}|^4/3$ on $\langle 111\rangle$. The two directions therefore agree on $m^*$ exactly and first differ at quartic order, by $\frac{2}{3}A\,|\mathbf{p}|^4$. Practical consequence: measuring both legs is not new physics but a *fit-window diagnostic* — if quadratic fits along $\Gamma \to X$ and $\Gamma \to L$ over the same $|\mathbf{p}|$ window disagree beyond combined errors, quartic contamination is leaking in and the window must shrink.

**The degenerate contrast (why the hole is different).** The LiF valence-band top at $\Gamma$ is threefold degenerate (F-2$p$-like). For a degenerate level the scalar expansion (6) is replaced by a $3 \times 3$ matrix quadratic form (the Luttinger–Kohn $\mathbf{k}\cdot\mathbf{p}$ Hamiltonian) with *three* independent cubic invariants (Luttinger parameters $\gamma_1, \gamma_2, \gamma_3$): heavy and light branches, genuinely anisotropic and warped. This is precisely why the hole polaron needs a multi-band, matrix-product-weight treatment and stays deferred (methodology note, deferred-work section); nothing in this note's scalar analysis applies to it.

**Where the quasiparticle branch ends.** A polaron at momentum $\mathbf{p}$ can shed a phonon and recoil; the dispersion flattens as $E_0(\mathbf{p})$ approaches the continuum edge $E_0(\Gamma) + \omega_{\mathrm{LO}}$ ($\omega_{\mathrm{LO}} \approx 0.08$ eV in LiF), and beyond it there is no sharp branch to fit. All mass fits must stay well below this threshold. Two sampling caveats tied to $|\mathbf{p}|$: the projection gap that suppresses excited-state contamination in the fixed-$\tau$ estimator shrinks as the threshold is approached, so the required $\tau$ *grows* with $|\mathbf{p}|$ (hence a $\tau$-doubling guard at the largest fitted momentum); and at $\mathbf{p} \neq \Gamma$ the propagator phases are genuinely complex, so $\langle \mathrm{sign} \rangle$ decays with $|\mathbf{p}|$ and sets the largest reachable grid index in practice.

# 6 Reading

**Setyawan & Curtarolo (2010).** W. Setyawan, S. Curtarolo, *High-throughput electronic band structure calculations: Challenges and tools*, Comput. Mater. Sci. **49**, 299 (2010); arXiv:1004.2974. *The* reference for BZ shapes, point labels, and standard paths for every Bravais lattice; the fcc figure and coordinate table cover everything in Section 2. Short and practical — read this one first.

**Hinuma et al. (2017).** Y. Hinuma, G. Pizzi, Y. Kumagai, F. Oba, I. Tanaka, *Band structure diagram paths based on crystallography*, Comput. Mater. Sci. **128**, 140 (2017); arXiv:1602.06402. The modern, crystallographically careful successor (the SeeK-path paper). The companion web tool at materialscloud.org draws the LiF BZ with $\Gamma/X/L$ labeled interactively — the fastest way to build geometric intuition for Section 2.

**Yu & Cardona.** P. Y. Yu, M. Cardona, *Fundamentals of Semiconductors*, 4th ed., Springer (2010), ch. 2. The $\mathbf{k}\cdot\mathbf{p}$ method: why band extrema at high-symmetry points carry the mass-tensor structure their little group dictates, including the degenerate (Luttinger–Kohn) case of Section 5.

**Dresselhaus, Dresselhaus & Jorio.** *Group Theory: Application to the Physics of Condensed Matter*, Springer (2008). The full machinery — little groups, stars, representations at $\Gamma/X/L$, and **kp** at high-symmetry points. The reference to reach for when the hole polaron (degenerate $\Gamma$ level) is picked up.

**Ashcroft & Mermin.** *Solid State Physics*, Saunders (1976), chs. 4–5 and 8. Background on the reciprocal lattice, Brillouin zones, and Bloch bands (Section 1), if a first-principles refresher is wanted.